

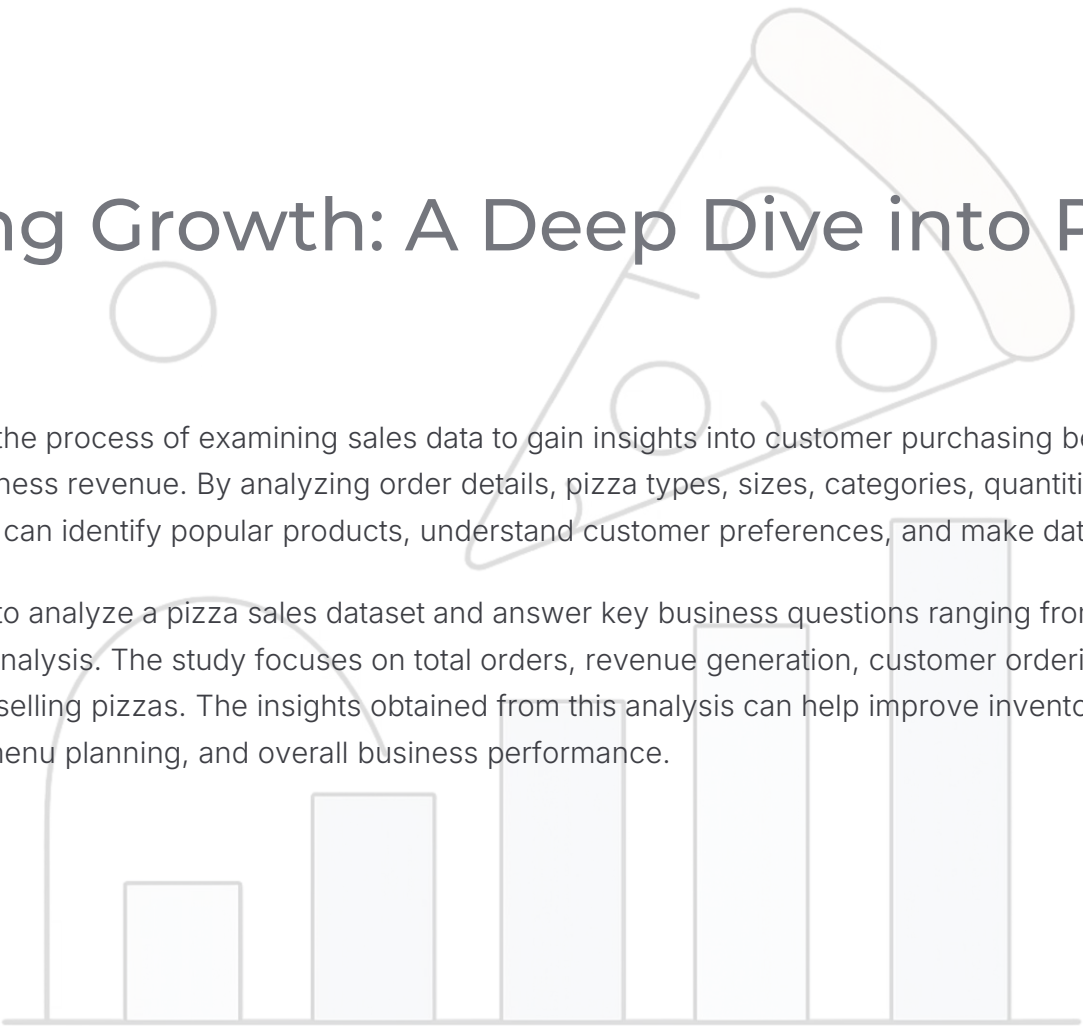


Pizza Sales Analysis

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Unlocking Growth: A Deep Dive into Pizza Sales



Pizza sales analysis is the process of examining sales data to gain insights into customer purchasing behavior, product performance, and business revenue. By analyzing order details, pizza types, sizes, categories, quantities sold, and revenue generated, businesses can identify popular products, understand customer preferences, and make data-driven decisions.

This project uses SQL to analyze a pizza sales dataset and answer key business questions ranging from basic sales metrics to advanced revenue analysis. The study focuses on total orders, revenue generation, customer ordering patterns, category performance, and top-selling pizzas. The insights obtained from this analysis can help improve inventory management, marketing strategies, menu planning, and overall business performance.

Objectives of This Analysis

This analysis aims to provide a comprehensive understanding of pizza sales performance, identifying key trends and actionable strategies for operational enhancement and revenue growth.



Core Goals

- Calculate total orders and total revenue.
- Identify the most popular pizza sizes and types.
- Analyze sales performance by category.
- Understand customer ordering patterns by time and date.
- Determine top-performing pizzas based on quantity and revenue.
- Analyze revenue contribution and cumulative revenue trends.
- Generate business insights to support decision-making.



BEGINNER

Foundational Pizza Industry Questions

Questions 1-5



Q1: Retrieve the total number of orders placed.

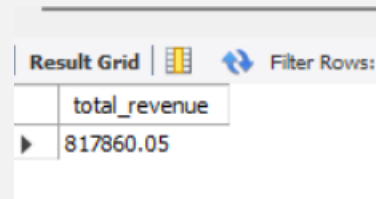
```
SELECT  
    COUNT(order_id) AS total_orders  
FROM  
    orders;
```

Result Grid	
	total_orders
▶	21350

This query calculates the **total number of orders placed by customers** in the pizza store. It counts all order records available in the `orders` table and returns the result as **total_orders**.

Q2: Calculate the total revenue generated from pizza sales.

```
SELECT  
    ROUND(SUM(p.price * od.quantity), 2) AS total_revenue  
FROM  
    order_details od  
    INNER JOIN  
    pizzas AS p ON p.pizza_id = od.pizza_id;
```



The screenshot shows a 'Result Grid' interface. It has a header row with 'total_revenue' and a single data row below it containing the value '817860.05'. There are icons for 'Filter Rows' and a refresh button.

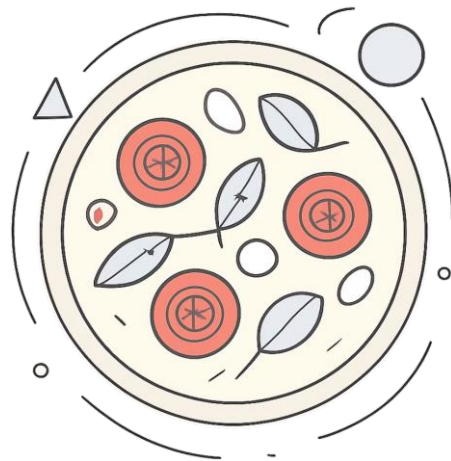
total_revenue
817860.05

This query calculates the **total revenue generated from pizza sales**. It multiplies the price of each pizza by the quantity sold and then sums all sales amounts to determine the overall revenue. The result is rounded to two decimal places.

Q3: Identify the highest-priced pizza.

```
SELECT  
  p.price AS highest_price, pt.name  
FROM  
  pizzas AS p  
  INNER JOIN  
  pizza_types AS pt ON pt.pizza_type_id = p.pizza_type_id  
ORDER BY p.price DESC  
LIMIT 1;
```

Result Grid			Filter Rows:
	highest_price	name	
▶	35.95	The Greek Pizza	



This query identifies the **highest-priced pizza** available in the menu. It retrieves the pizza name and its price, sorts all pizzas in descending order based on price, and returns only the most expensive pizza.

Q4: Identify the most common pizza size ordered.

```
SELECT  
  COUNT(od.order_details_id) AS order_count, p.size  
FROM  
  order_details AS od  
  INNER JOIN  
  pizzas AS p ON p.pizza_id = od.pizza_id  
GROUP BY p.size  
ORDER BY order_count desc  
LIMIT 1;
```

Result Grid		
	order_count	size
▶	18526	L

This query identifies the **most commonly ordered pizza size** by counting the number of orders for each pizza size and returning the size with the highest order count.



Q5: List the top 5 most ordered pizza types along with their quantities.



SQL Query

```
SELECT
    SUM(od.quantity) AS quantity, pt.name
FROM
    order_details AS od
    INNER JOIN
    pizzas AS p ON p.pizza_id = od.pizza_id
    INNER JOIN
    pizza_types AS pt ON pt.pizza_type_id = p.pizza_type_id
GROUP BY pt.name
ORDER BY quantity DESC
LIMIT 5
```

Result Grid			Filter Rows:
	quantity	name	
▶	2453	The Classic Deluxe Pizza	
	2432	The Barbecue Chicken Pizza	
	2422	The Hawaiian Pizza	
	2418	The Pepperoni Pizza	
	2371	The Thai Chicken Pizza	

This query identifies the **Top 5 most ordered pizza types** based on the total quantity sold. It calculates the total number of pizzas sold for each pizza type and displays the five pizzas with the highest sales volume.

INTERMEDIATE

Strategic Pizza Business Questions

Questions 6-10

Q6: Join the necessary tables to find the total quantity of each pizza category ordered.

THE PATH TO PROFITABILITY

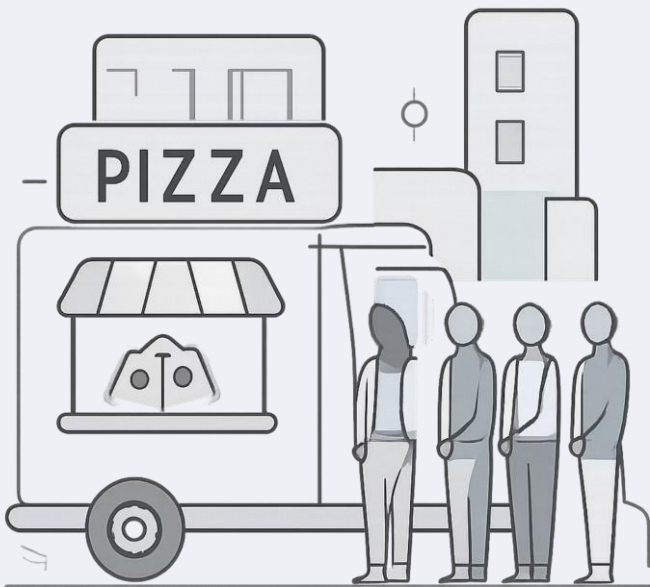


SQL query

```
SELECT
    SUM(od.quantity) AS quantity, pt.category
FROM
    order_details AS od
    JOIN
    pizzas AS p ON p.pizza_id = od.pizza_id
    JOIN
    pizza_types AS pt ON pt.pizza_type_id = p.pizza_type_id
GROUP BY pt.category;
```

	quantity	category
▶	14888	Classic
	11649	Veggie
	11987	Supreme
	11050	Chicken

This query calculates the **total quantity of pizzas ordered for each pizza category**. It groups pizzas by category (such as Classic, Chicken, Supreme, and Veggie) and sums the quantities sold to determine the popularity of each category.



Q7: Determine the distribution of orders by hour of the day.

```
SELECT
    COUNT(o.order_id) order_count,
    HOUR(o.order_time) hours
FROM
    orders o
GROUP BY HOUR(o.order_time);
```

Result Grid			 Filter Rows:
	order_count	hours	
	2455	13	
	1472	14	
	1468	15	
	1920	16	
	2336	17	
	2399	18	
	2009	19	
	1642	20	
	1198	21	
	663	22	
	28	23	
	8	10	
	1	9	

This query analyzes the **distribution of orders by hour of the day**. It counts how many orders were placed during each hour and helps identify peak business hours when customer demand is highest.

Q8: Group the orders by date and calculate the average number of pizzas ordered per day.



SQL query

```
SELECT  
  ROUND(AVG(quantity), 0) avg_quantity  
FROM  
  (SELECT  
    SUM(od.quantity) quantity, order_date  
  FROM  
    order_details od  
  JOIN orders o ON o.order_id = od.order_id  
  GROUP BY o.order_date) AS order_quantity;
```

Result Grid	
	avg_quantity
▶	138

This query calculates the **average number of pizzas ordered per day**. First, it determines the total quantity of pizzas sold each day, and then it calculates the average of those daily totals.

Q9: Join relevant tables to find the category-wise distribution of pizzas.

```
SELECT  
  COUNT(p.pizza_id) pizza_count, pt.category  
FROM  
  pizzas p  
  INNER JOIN  
  pizza_types AS pt ON pt.pizza_type_id =  
  p.pizza_type_id  
GROUP BY pt.category;
```

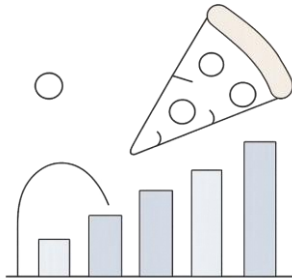
Result Grid			Filter Rows:
	pizza_count	category	
▶	18	Chicken	
	26	Classic	
	25	Supreme	
	27	Veggie	



This query calculates the **distribution of pizzas across different categories**. It counts how many pizza varieties belong to each category, such as Classic, Chicken, Supreme, and Veggie.

Q10: Determine the top 3 most ordered pizza types based on revenue.

THE PATH TO PROFITABILITY



SQL query

```
SSELECT
  ROUND(SUM(p.price * od.quantity), 2) AS revenue, pt.name
FROM
  order_details od
  JOIN
  pizzas AS p ON p.pizza_id = od.pizza_id
  JOIN
  pizza_types AS pt ON pt.pizza_type_id = p.pizza_type_id
GROUP BY pt.name
ORDER BY revenue DESC
LIMIT 3
```

	revenue	name
▶	43434.25	The Thai Chicken Pizza
	42768	The Barbecue Chicken Pizza
	41409.5	The California Chicken Pizza

This query identifies the **Top 3 pizza types based on revenue generated**. It calculates the revenue earned by each pizza type, show them from highest to lowest revenue, and returns the top three performers.

ADVANCED

ADVANCED LEVEL - Complex Pizza Industry Analysis

Questions 11-13

Q11: Calculate the percentage contribution of each pizza type to total revenue.

```
SELECT  
  ROUND(((ROUND(SUM(od.quantity * p.price), 2) / (SELECT  
    ROUND(SUM(od.quantity * p.price), 2)  
  FROM  
    order_details AS od  
    JOIN  
      pizzas AS p ON p.pizza_id = od.pizza_id)) * 100),  
  2) AS total_revenue,  
  pt.category  
FROM  
  order_details AS od  
  JOIN  
    pizzas AS p ON p.pizza_id = od.pizza_id  
  JOIN  
    pizza_types AS pt ON pt.pizza_type_id = p.pizza_type_id  
GROUP BY pt.category  
ORDER BY total_revenue DESC;
```

	total_revenue	category
▶	26.91	Classic
	25.46	Supreme
	23.96	Chicken
	23.68	Veggie



This query calculates the **percentage contribution of each pizza category to total revenue**. It determines how much revenue each category generates and expresses it as a percentage of the overall revenue.

Q12: Analyze the cumulative revenue generated over time.

```
SELECT *,  
    ROUND(SUM(revenue) OVER (ORDER BY order_date), 2) AS  
    cumulative_revenue
```

```
FROM
```

```
(
```

```
    SELECT
```

```
        o.order_date,
```

```
        ROUND(SUM(od.quantity * p.price), 2) AS revenue
```

```
    FROM orders AS o
```

```
    JOIN order_details od
```

```
        ON od.order_id = o.order_id
```

```
    JOIN pizzas p
```

```
        ON p.pizza_id = od.pizza_id
```

```
    GROUP BY order_date
```

```
) AS per_day_revenue;
```

	order_date	revenue	cumulative_revenue
▶	2015-01-01	2713.85	2713.85
	2015-01-02	2731.9	5445.75
	2015-01-03	2662.4	8108.15
	2015-01-04	1755.45	9863.6
	2015-01-05	2065.95	11929.55
	2015-01-06	2428.95	14358.5
	2015-01-07	2202.2	16560.7



This query analyzes the **cumulative revenue generated over time**. It first calculates the revenue earned each day and then continuously adds each day's revenue to the previous total, showing how total revenue grows over time.

Q13: Determine the top 3 most ordered pizza types based on revenue for each pizza category.

```
SELECT *
FROM
(
  SELECT
    category,
    name,
    revenue,
    RANK() OVER (
      PARTITION BY category
      ORDER BY revenue DESC
    ) AS result
  FROM
  (
    SELECT
      SUM(od.quantity * p.price) AS revenue,
      pt.name,
      pt.category
    FROM order_details od
    JOIN pizzas p
      ON p.pizza_id = od.pizza_id
    JOIN pizza_types pt
      ON pt.pizza_type_id = p.pizza_type_id
    GROUP BY pt.name, pt.category
  ) AS raw_data
  ) AS rank_data
WHERE result <= 3;
```

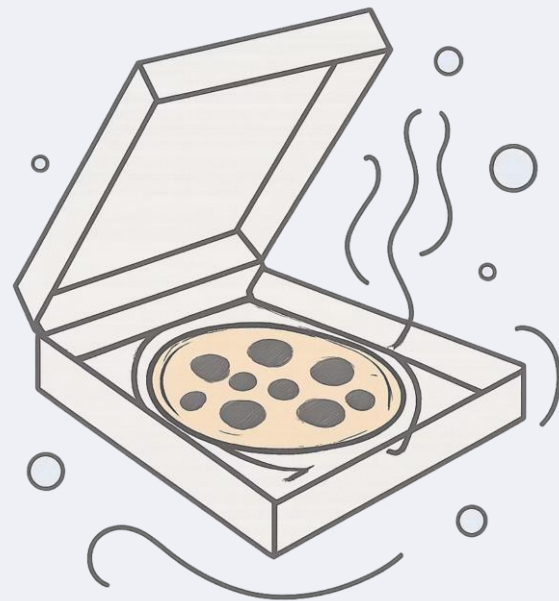
	category	name	revenue	result
►	Chicken	The Thai Chicken Pizza	43434.25	1
	Chicken	The Barbecue Chicken Pizza	42768	2
	Chicken	The California Chicken Pizza	41409.5	3
	Classic	The Classic Deluxe Pizza	38180.5	1
	Classic	The Hawaiian Pizza	32273.25	2
	Classic	The Pepperoni Pizza	30161.75	3
	Supreme	The Spicy Italian Pizza	34831.25	1
	Supreme	The Italian Supreme Pizza	33476.75	2
	Supreme	The Sicilian Pizza	30940.5	3
	Veggie	The Four Cheese Pizza	32265.70000000065	1
	Veggie	The Mexicana Pizza	26780.75	2
	Veggie	The Five Cheese Pizza	26066.5	3

This query identifies the **Top 3 pizza types by revenue within each pizza category**. It calculates revenue for every pizza, ranks pizzas within their respective categories, and returns the three highest revenue-generating pizzas from each category.



Key Takeaways: SQL-Driven Insights

- Large-sized pizzas were the most frequently ordered, indicating a strong customer preference for larger portions.
- A few pizza types generated the highest sales and revenue, making them key products for the business.
- Customer demand peaked during lunch and dinner hours, highlighting important periods for staffing and inventory planning.
- The Classic category contributed the most to overall sales performance, demonstrating its popularity among customers.
- SQL techniques such as JOINS, Aggregate Functions, Subqueries, and Window Functions transformed raw sales data into actionable business insights for decision-making.



Recommendation/Suggestion

- **Focus on top-performing pizzas** by promoting the highest-selling and highest-revenue pizza types through special offers and combo deals.
- **Maintain adequate inventory** for popular pizza sizes and categories, especially large pizzas and the Classic category, to avoid stock shortages.
- **Optimize staffing during peak hours** (lunch and dinner periods) to improve customer service and reduce waiting times.
- **Review low-performing pizzas** and consider improving recipes, pricing, or promotional strategies to increase their sales.
- **Use sales insights for future planning** by regularly analyzing customer preferences and revenue trends to support data-driven business decisions.



Transforming Pizza Business with Data Insights

Through SQL-based analysis, valuable insights were extracted from pizza sales data regarding customer preferences, sales performance, revenue trends, and product profitability. These insights can support data-driven decision-making, improve operational efficiency, optimize menu offerings, and enhance overall business performance.





Thank You!